



SILENT SENTINEL

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G5 Thermal Camera External Universal Protocol

v2.4.4

1 Overview

This document describes **External Universal Protocol** to control **G5** thermal block cameras (combined lens and camera). This document lists supported commands by **G5Camera** driver. In the **NexOS** systems the **G5** thermal camera always has name **Camera1**.

2 Lens control messages

The **Lens** submodule is part of the **Camera1** module. This submodule is responsible for lens control.

Table 2.1 - Action Commands and Parameter Change Commands: Messages begin with **.../Command/Camera1/Lens/...** If the command is successfully sent to the lens controller, the system responds with **Ack**. Expected response time: **ip ping time to the system + 5 msec**. If a particular command is not supported by **G5Camera** driver the **NexOS** system will return **Ack** but the action command or parameter will not be accepted by lens driver.

Command	Description
ZoomIn	Move zoom tele (in).
ZoomOut	Move zoom wide (out).
ZoomToPos:n	Move zoom to position. Parameter n : integer position value from 0 (full wide) to 65535 (full tele).
ZoomPos:n	Move zoom to position. Parameter n : integer position value from 0 (full wide) to 65535 (full tele).
ZoomHwPos:n	Move zoom to hardware position. Parameter n : integer position value from 0 (full wide) to 1000 (full tele).
ZoomStop	Stop zoom movement including stop zoom to position command.
FocusFar	Move focus far.
FocusNear	Move focus near.

Command	Description
FocusToPos:n	Move focus to position. Parameter n : integer position value from 0 (full near) to 65535 (full far).
FocusPos:n	Move focus to position. Parameter n : integer position value from 0 (full near) to 65535 (full far).
FocusHwPos:n	Move focus to hardware position. Parameter n : integer hardware position value in range [0-1000].
IrisOpen	Not supported by G5Camera driver.
IrisClose	Not supported by G5Camera driver.
IrisToPos:n	Not supported by G5Camera driver.
IrisPos:n	Not supported by G5Camera driver.
IrisHwPos:n	Not supported by G5Camera driver.
IrisStop	Not supported by G5Camera driver.
RunAf	Start autofocus. It uses built in auto focus algorithm.
StopAf	Not supported by G5Camera driver.
Restart	Not supported by G5Camera driver.
FocusMode:n	Stored locally. No hardware effect. G5 always operates in push focus mode regardless of this value. Accepted values: 0 - push, 1 - auto, 2 - other.
FilterMode:n	Not supported by G5Camera driver.
AfRoiMode:n	Not supported by G5Camera driver.
ZoomSpeed:n	Set zoom speed. Parameter n : integer speed value [1-100%] of max hardware zoom speed.
FocusSpeed:n	Set focus speed. Parameter n : integer speed value [1-100%] of max hardware focus speed.
IrisSpeed:n	Not supported by G5Camera driver.
FocusFactorThreshold:n	Not supported by G5Camera driver.
RefocusTimeoutSec:n	Not supported by G5Camera driver.
IrisMode:n	Not supported by G5Camera driver.
ExtenderMode:n	Not supported by G5Camera driver.
StabiliserMode:n	Not supported by G5Camera driver.
AfRange:n	Not supported by G5Camera driver.
XFovDeg	Set horizontal field of view angle, degree. Parameter n : float horizontal field of view angle [deg].

Command	Description
YFovDeg:n	Set vertical field of view angle, degree. Parameter n : float vertical field of view angle [deg]. G5Camera converts Y FOV to X FOV using FullHD aspect ratio (factor 1.757) and sends the resulting X FOV command to the lens.
Custom1:n	Not supported by G5Camera driver.
Custom2:n	Not supported by G5Camera driver.
Custom3:n	Not supported by G5Camera driver.

Table 2.2 - Lens submodule queries. Messages start from **.../Query/Camera1/Lens/...** Response from the system starts with **.../Reply/Camera1/Lens/...** Expected response time: **ip ping time to the system + 5 msec**. If a particular query is not supported by **G5Camera** driver the **NexOS** system will return default value which is not related to real behavior of lens.

Command	Description
ZoomPos	Query zoom position. Response: .../ZoomPos:n . Parameter n : integer position value from 0 (full wide) to 65535 (full tele).
ZoomHwPos	Query zoom hardware position. Response: .../ZoomHwPos:n . Parameter n : integer position value from 0 (full wide) to 1000 (full tele).
FocusPos	Query focus position. Response: .../FocusPos:n . Parameter n : position value from 0 (full near) to 65535 (full far).
FocusHwPos	Query focus hardware position. Response: .../FocusHwPos:n . Parameter n : integer hardware position value.
IrisPos	Not supported by G5Camera driver.
IrisHwPos	Not supported by G5Camera driver.
ZoomSpeed	Query zoom speed. Response: .../ZoomSpeed:n . Parameter n : speed value [1-100%] (relative for particular lens).
FocusSpeed	Query focus speed. Response: .../FocusSpeed:n . Parameter n : speed value [1-100%] (relative for particular lens).
IrisSpeed	Not supported by G5Camera driver.
FocusMode	Returns locally stored focus mode value. Response: .../FocusMode:n . Default: 0 (push focus).
FilterMode	Not supported by G5Camera driver.
AfRoiMode	Not supported by G5Camera driver.
IsConnected	Query connection status. Connection status show if lens controller has communication with lens hardware. Response: .../IsConnected:n . Parameter n : 0 - no response from lens, 1 - connected.
IsOpen	Query lens controller initialization status. Response: .../IsOpen:n . Parameter n (integer): 0 - lens controller not init, 1 - init.

Command	Description
FocusFactor	Not supported by G5Camera driver.
FocusFactorThreshold	Not supported by G5Camera driver.
RefocusTimeoutSec	Not supported by G5Camera driver.
IsAfActive	Not supported by G5Camera driver.
IrisMode	Not supported by G5Camera driver.
ExtenderMode	Not supported by G5Camera driver.
AfRange	Not supported by G5Camera driver.
XFovDeg	Query current horizontal field of view angle, degree. Response: .../FovHorizontal:n . Parameter n : float horizontal field of view angle [deg].
YFovDeg	Query current vertical field of view angle, degree. Response: .../FovVertical:n . Parameter n : float vertical field of view angle [deg]. The G5 camera only returns horizontal FOV. Note : the vertical FOV is calculated by G5Camera driver according to FullHD video aspect ratio.
Temperature	Not supported by G5Camera driver.
Custom1	Not supported by G5Camera driver.
Custom2	Not supported by G5Camera driver.
Custom3	Not supported by G5Camera driver.

Some example messages:

[client -> system] **ID:14/Command/Camera1/Lens/ZoomToPos:12000** - Go to zoom position 12000.

[system -> client] **ID:14/Ack** - Command accepted.

[client -> system] **ID:16/Query/Camera1/Lens/FocusPos** - Query focus position from camera 1.

[system -> client] **ID:16/Reply/Camera1/Lens/FocusPos:100** - Focus position from camera 1.

3 Camera control messages

Camera is a submodule which is part of **Camera1** module. **Camera** submodule is responsible for camera core (image sensor) control.

Table 3.1 - Action commands and commands to change parameters. Messages start from **.../Command/Camera1/Camera/....** Response from the system: **Ack**. Expected response time: **ip ping time to the system + 5 msec**. If a particular command is not supported by **G5Camera** driver the **NexOS** system will return **Ack** but the action command or parameter will not be accepted by camera driver.

Command	Description
Restart	Restart camera controller.
Nuc	Do NUC (Flat Field Correction).

Command	Description
ApplyParams	Not supported by G5Camera driver.
SaveParams	Not supported by G5Camera driver.
MenuOn	Not supported by G5Camera driver.
MenuOff	Not supported by G5Camera driver.
MenuSet	Not supported by G5Camera driver.
MenuUp	Not supported by G5Camera driver.
MenuDown	Not supported by G5Camera driver.
MenuLeft	Not supported by G5Camera driver.
MenuRight	Not supported by G5Camera driver.
FreezeSec:n	Freeze image for n seconds. 0 - freeze without timeout (integer). The G5 camera doesn't support timeout.
Defreeze	Disable freeze.
DisplayMode:n	Set display (OSD) mode. Parameter n (integer): 0 - OSD off, 1 - OSD on.
ExposureMode:n	Not supported by G5Camera driver.
ExposureTime:n	Not supported by G5Camera driver.
WhiteBalanceMode:n	Not supported by G5Camera driver.
WhiteBalanceArea:n	Not supported by G5Camera driver.
WideDynamicRangeMode:n	Not supported by G5Camera driver.
StabilisationMode:n	Set image stabilization mode. Parameter n inside camera core (integer): 0 - Off, 1 - On.
IsoSensitivity:n	Not supported by G5Camera driver.
SceneMode:n	AGC ROI mode. Parameter n (integer): 1 - ¼ of image. 2 - ½ of image 3 - ¾ of image 4 - whole image 5 - top 1/3 of image 6 - middle 1/3 of image 7 - bottom 1/3 of image.
BrightnessMode:n	Not supported by G5Camera driver.
Brightness:n	Not supported by G5Camera driver.

Command	Description
Contrast:n	Set local area contrast enhancement mode. Parameter n (integer): 0 - Off. 1 - Low. 2 - Medium. 3 - High.
GainMode:n	Gain mode. Parameter n (integer): 0 - Turn off Auto Gain Control. 1 - Set the Auto Gain Control to Normal mode. 2 - Set the Auto Gain Control to Linear mode.
Gain:n	Set gain level. Parameter n (integer): gain level [0-100%].
SharpeningMode:n	Turbulence mitigation sharpness mode. Parameter n (integer): 0 - Off. 1 - Minimum. 2 - Low. 3 - Medium. 4 - High. 5 - Maximum.
Sharpening:n	Not supported by G5Camera driver.
Palette:n	Set palette. Parameter n (integer): palette mode 0-7.
AgcMode:n	Set analog gain control mode. Parameter n (integer): 0 - Turn off Auto Gain Control. 1 - Set the Auto Gain Control to Normal mode. 2 - Set the Auto Gain Control to Linear mode.
ShutterMode:n	Not supported by G5Camera driver.
ShutterPos:n	Not supported by G5Camera driver.
ShutterSpeed:n	Not supported by G5Camera driver.
ExposureCompensationMode:n	Not supported by G5Camera driver.
ExposureCompensationPos:n	Not supported by G5Camera driver.
DefogMode:n	Not supported by G5Camera driver.
DehazeMode:n	Dehaze mode (Turbulence mitigation). Parameter n (integer): 0 - Off. 1 - Minimum. 2 - Low. 3 - Medium. 4 - High. 5 - Maximum.
NoiseReductionMode:n	Not supported by G5Camera driver.

Command	Description
BlackAndWhiteFilterMode:n	Not supported by G5Camera driver.
FilterMode:n	Set filter mode. Parameter n (integer): 0 - Off. 1 - Make the image appear slightly sharper. 5 - Make the image appear very sharp with enhanced edges.
NucMode:n	Set NUC mode for thermal cameras. Parameter n (integer): 0 - Manual. 1 - Auto.
AutoNucIntervalSec:n	Set auto NUC interval for thermal cameras. Parameter n (integer): nuc interval [sec] 1 - 5000 . To disable auto NUC use NucMode:0 .
AutoNucIntervalMin:n	Set auto NUC interval for thermal cameras. Parameter n (integer): nuc interval [minutes] 1 - 83 . To disable auto NUC use NucMode:0 .
DdeMode:n	Not supported by G5Camera driver.
DdeLevel:n	Not supported by G5Camera driver.
AlcGate:n	Set analog gain and level time constant. Parameter n (integer): 0-10. Default 4 . Decreasing the value will cause the AGC to react quicker. A value of 1 tends to cause the image to flicker occasionally.
Sensitivity:n	Set sensor sensitivity. Parameter n (integer): sensitivity. Values: -1, 0, 1.
ChangingMode:n	Not supported by G5Camera driver.
ChangingLevel:n	Not supported by G5Camera driver.
ChromaLevel:n	Not supported by G5Camera driver.
Detail:n	Not supported by G5Camera driver.
Profile:n	Not supported by G5Camera driver.
Custom1:n	Turbulence mitigation ROI. Parameter n (integer): 1 - ¼ of image 2 - ½ of image 3 - ¾ of image 4 - whole image 5 - top 1/3 of image 6 - middle 1/3 of image 7 - bottom 1/3 of image.

Command	Description
Custom2:n	Turbulence mitigation quality mode. Parameter n (integer). Default value 2 (40%): 0 - 0 (%). 1 - 20 (%). 2 - 40 (%). 3 - 60 (%). 4 - 80 (%). 5 - 100 (%).
Custom3:n	Turbulence mitigation level. Parameter n (integer). Default value 0 (100%): 0 - 100 (%). 1 - 80 (%). 2 - 60 (%). 3 - 40 (%). 4 - 20 (%). 5 - 0 (%).
AgcRoiX0:n	Set AGC ROI left-top X coordinate. Parameter n (integer): 0-2048. The ROI is sent to hardware only when all four coordinates (X0, Y0, X1, Y1) have been set.
AgcRoiY0:n	Set AGC ROI left-top Y coordinate. Parameter n (integer): 0-2048. The ROI is sent to hardware only when all four coordinates (X0, Y0, X1, Y1) have been set.
AgcRoiX1:n	Set AGC ROI right-bottom X coordinate. Parameter n (integer): 0-2048. The ROI is sent to hardware only when all four coordinates (X0, Y0, X1, Y1) have been set.
AgcRoiY1:n	Set AGC ROI right-bottom Y coordinate. Parameter n (integer): 0-2048. The ROI is sent to hardware only when all four coordinates (X0, Y0, X1, Y1) have been set.
Type	Query camera type. Response: .../Type:string . Parameter string (string): camera type identifier: G5 .

Table 3.2 - Camera submodule queries. Messages start from **.../Query/Camera1/Camera/...** Response from the system starts with **.../Reply/Camera1/Camera/...** Expected response time: **ip ping time to the system + 5 msec**. If a particular query is not supported by **G5Camera** driver the **NexOS** system will return default value which is not related to real behavior of camera.

Command	Description
DisplayMode	Returns last set display (OSD) mode. Response: .../DisplayMode:n . Parameter n (integer): 0 - OSD off, 1 - OSD on.
ExposureMode	Not supported by G5Camera driver.
ExposureTime	Not supported by G5Camera driver.

Command	Description
WhiteBalanceMode	Not supported by G5Camera driver.
WhiteBalanceArea	Not supported by G5Camera driver.
WideDynamicRangeMode	Not supported by G5Camera driver.
StabilisationMode	Query image stabilization mode inside camera core. Response: .../StabilisationMode:n . Parameter n inside camera core (integer): 0 - Off, 1 - On.
IsoSensitivity	Not supported by G5Camera driver.
SceneMode	Query AGC ROI mode. Response: .../SceneMode:n . Parameter n (integer): 1 - ¼ of image. 2 - ½ of image 3 - ¾ of image 4 - whole image 5 - top 1/3 of image 6 - middle 1/3 of image 7 - bottom 1/3 of image.
BrightnessMode	Not supported by G5Camera driver.
Brightness	Not supported by G5Camera driver.
Contrast	Query area contrast enhancement mode. Response: .../Contrast:n . Parameter n (integer): 0 - Off. 1 - Low. 2 - Medium. 3 - High.
GainMode	Query gain mode. Response: .../GainMode:n . Parameter n (integer): 0 - Turn off Auto Gain Control. 1 - Set the Auto Gain Control to Normal mode. 2 - Set the Auto Gain Control to Linear mode.
Gain	Query gain level. Response: .../Gain:n . Parameter n (integer): gain level [0-100%].
SharpeningMode	Query turbulence mitigation sharpness mode. Response: .../SharpeningMode:n . Parameter n (integer): 0 - Off. 1 - Minimum. 2 - Low. 3 - Medium. 4 - High. 5 - Maximum.
Sharpening	Not supported by G5Camera driver.

Command	Description
Palette	Query palette. Response: .../Palette:n . Parameter n (integer): palette mode 0-7.
AgcMode	Query analog gain control mode. Response: .../AgcMode:n . Parameter n (integer): 0 - Turn off Auto Gain Control. 1 - Set the Auto Gain Control to Normal mode. 2 - Set the Auto Gain Control to Linear mode.
ShutterMode	Not supported by G5Camera driver.
ShutterPos	Not supported by G5Camera driver.
ShutterSpeed	Not supported by G5Camera driver.
ExposureCompensationMode	Not supported by G5Camera driver.
ExposureCompensationPos	Not supported by G5Camera driver.
DefogMode	Not supported by G5Camera driver.
DehazeMode	Query dehaze mode (Turbulence mitigation). Response: .../DehazeMode:n . Parameter n (integer): 0 - Off. 1 - Minimum. 2 - Low. 3 - Medium. 4 - High. 5 - Maximum.
NoiseReductionMode	Not supported by G5Camera driver.
BlackAndWhiteFilterMode	Not supported by G5Camera driver.
FilterMode	Query filter mode. Response: .../FilterMode:n . Parameter n (integer): 0 - Off. 1 - Make the image appear slightly sharper. 5 - Make the image appear very sharp with enhanced edges.
NucMode	Query NUC mode. Parameter n (integer): 0 - Manual. 1 - Auto.
AutoNucIntervalSec	Query auto NUC interval. Response: .../AutoNucIntervalSec:n . Parameter n (integer): nuc interval [sec] 0 (Off) - 5000 .
AutoNucIntervalMin	Query auto NUC interval. Response: .../AutoNucIntervalMin:n . Parameter n (integer): nuc interval [minutes] 0 (Off) - 83 .
DdeMode	Not supported by G5Camera driver.
DdeLevel	Not supported by G5Camera driver.

Command	Description
Temperature	Not supported by G5Camera driver.
AlcGate	Get analog gain and level time constant. Response: .../AlcGate:n . Parameter n (integer): 0-10. Default 4 . Decreasing the value will cause the AGC to react quicker. A value of 1 tends to cause the image to flicker occasionally.
Sensitivity	Query sensor sensitivity. Response: .../Sensitivity:n . Parameter n (integer): sensitivity. Values: -1, 0, 1.
ChangingMode	Not supported by G5Camera driver.
ChangingLevel	Not supported by G5Camera driver.
ChromaLevel	Not supported by G5Camera driver.
Detail	Not supported by G5Camera driver.
Profile	Not supported by G5Camera driver.
IsConnected	Query connection status. Response: .../IsConnected:n . Parameter n (integer): 0 - no responses from camera, 1 - connected. Shows if we have a response from the camera.
IsOpen	Query open status. Response: .../IsOpen:n . Parameter n (integer): 1 - open, 0 - not open. Open status shows if the camera controller is initialized or not but doesn't show if camera controller has communication with camera equipment.
Custom1	Query turbulence mitigation ROI. Response: .../Custom1:n . Parameter n (integer): 1 - ¼ of image 2 - ½ of image 3 - ¾ of image 4 - whole image 5 - top 1/3 of image 6 - middle 1/3 of image 7 - bottom 1/3 of image.
Custom2	Query turbulence mitigation quality mode. Response: .../Custom2:n . Parameter n (integer): 0 - 0 (%). 1 - 20 (%). 2 - 40 (%). 3 - 60 (%). 4 - 80 (%). 5 - 100 (%).

Command	Description
Custom3	Query turbulence mitigation level. Response: .../Custom3:n . Parameter n (integer): 0 - 100 (%). 1 - 80 (%). 2 - 60 (%). 3 - 40 (%). 4 - 20 (%). 5 - 0 (%).
AgcRoiX0	Query AGC ROI left-top X coordinate. Response: .../AgcRoiX0:n . Parameter n (integer): 0-2048.
AgcRoiY0	Query AGC ROI left-top Y coordinate. Response: .../AgcRoiY0:n . Parameter n (integer): 0-2048.
AgcRoiX1	Query AGC ROI right-bottom X coordinate. Response: .../AgcRoiX1:n . Parameter n (integer): 0-2048.
AgcRoiY1	Query AGC ROI right-bottom Y coordinate. Response: .../AgcRoiY1:n . Parameter n (integer): 0-2048.
Type	Query camera type. Response: .../Type:string . Parameter string (string): camera type identifier: G5, G5Atcom .
CoolerHours	Query thermal camera cooler working hours counter. Response: .../CoolerHours:n . Parameter n (integer): cooler hours counter.

Some example messages:

[client -> system] **ID:18/Command/Camera1/Camera/Sensitivity:1** - Set sensitivity.

[system -> client] **ID:18/Ack** - command accepted.

[client -> system] **ID:19/Command/Camera1/Camera/DehazeMode:1** - Set turbulence mitigation mode.

[system -> client] **ID:19/Ack** - command accepted.

[client -> system] **ID:21/Query/Camera1/Camera/IsConnected** - Query connection status.

[system -> client] **ID:21/Reply/Camera1/Camera/IsConnected:1** - Connected.